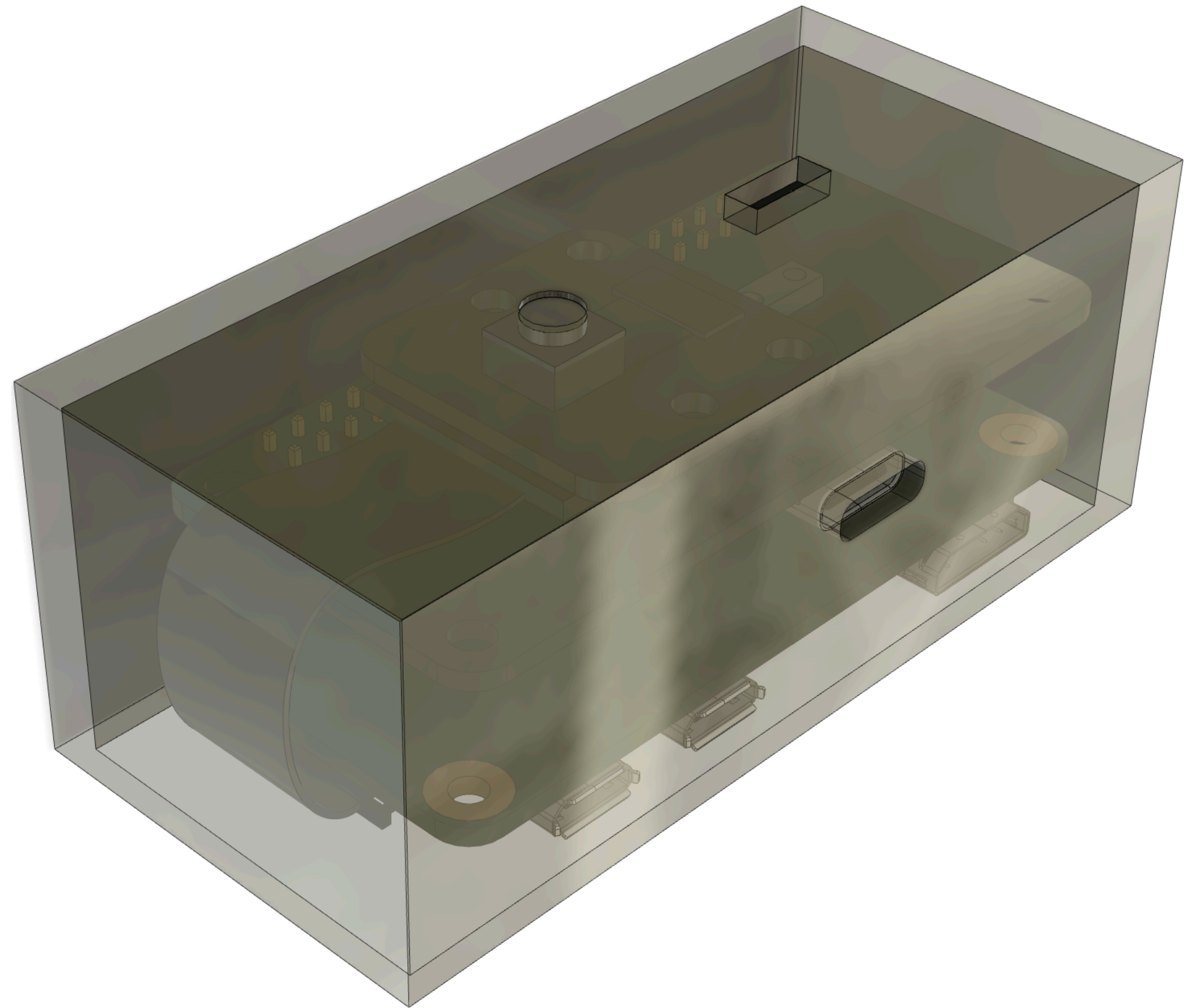
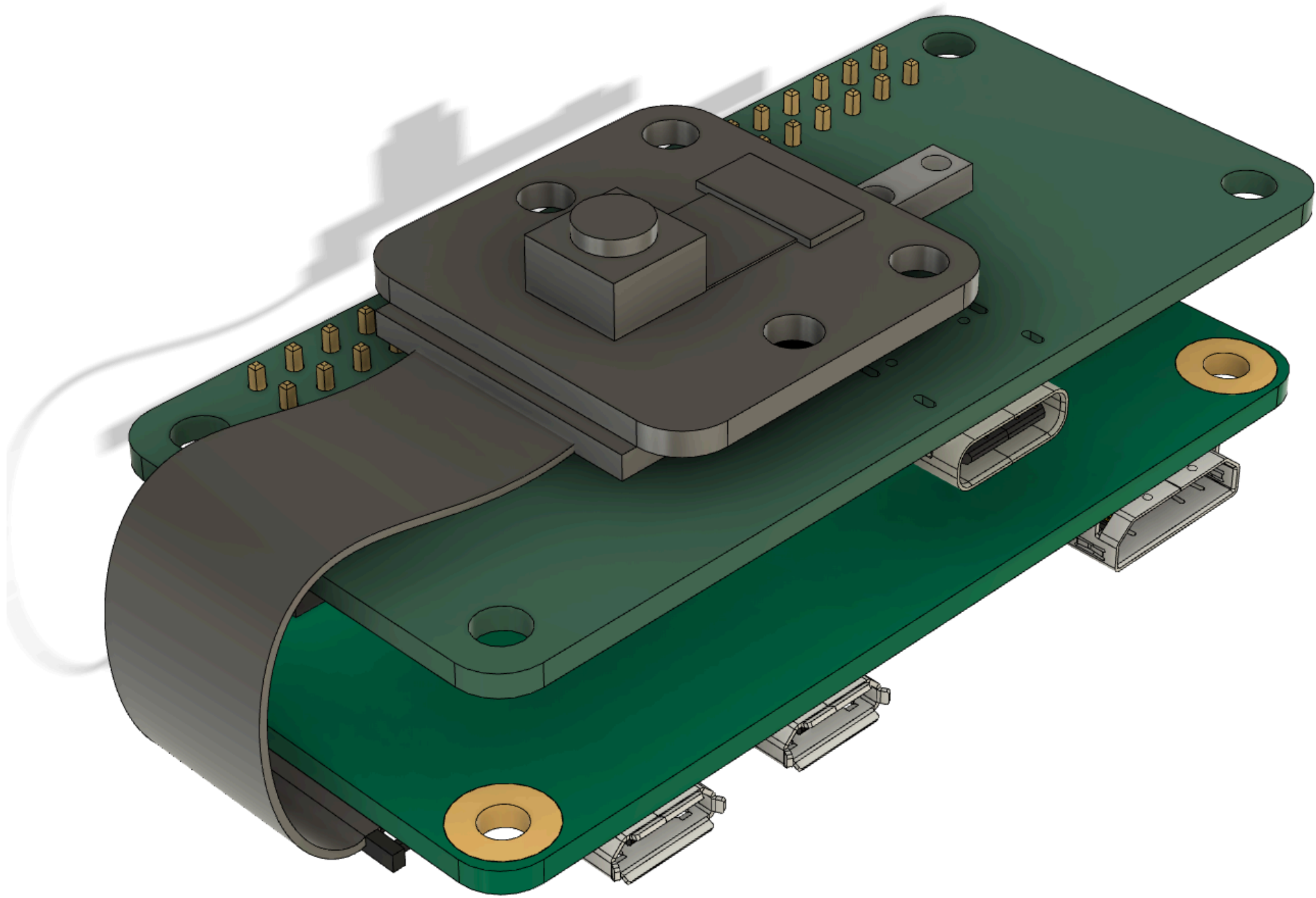


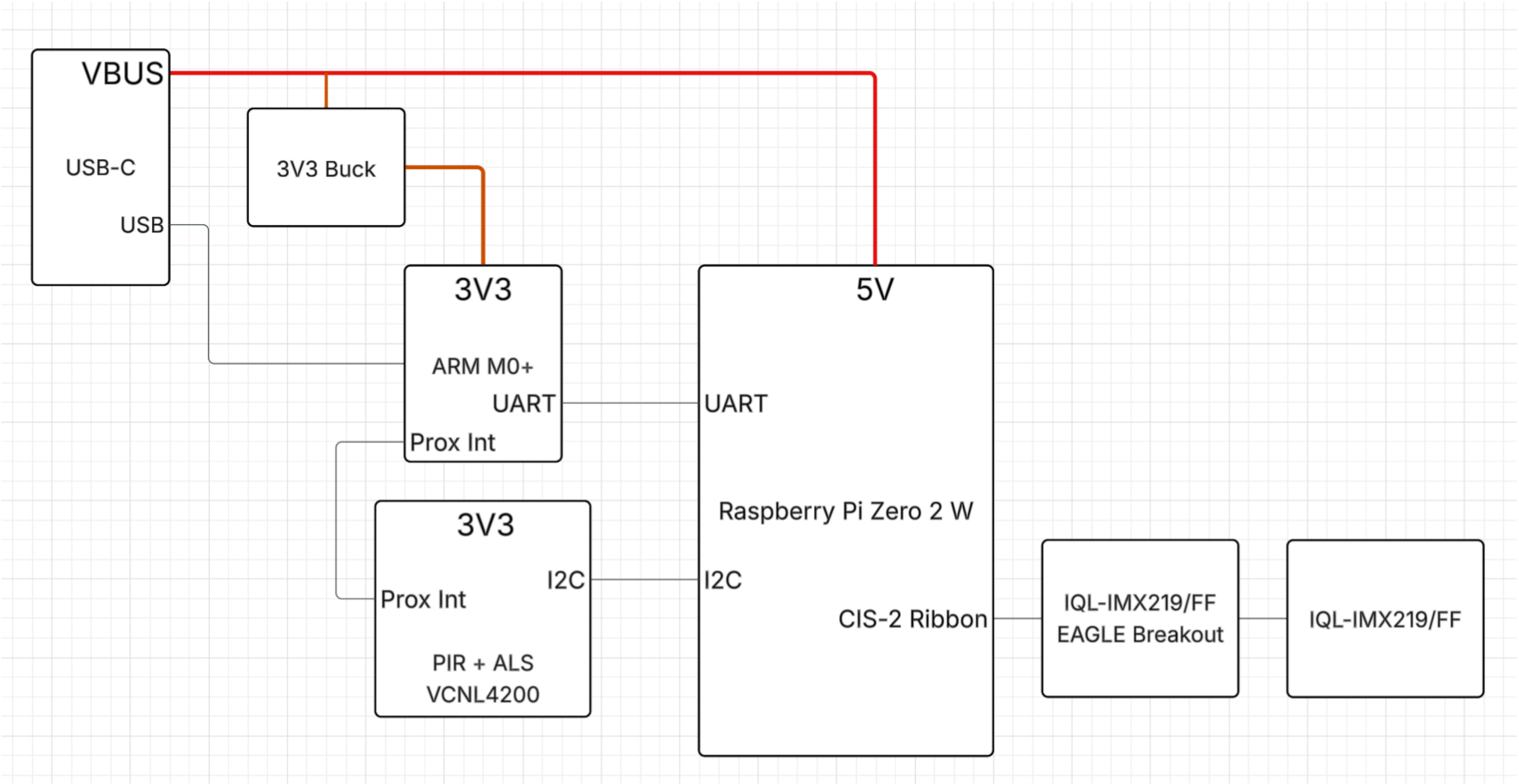
Tools For Humanity Camera Hat Case Study

Prepared by Oren Levy

Camera Hat | Intro & Renders



Camera Hat | Block Diagram



Camera Hat | Features

- Arm M0+ incorporated
 - Allows for 25mW standby consumption (Instead of 525mW) in conjunction with a load switch that disconnects the Raspberry Pi
 - When motion is detected, an interrupt wakes up the MCU, which closes the load switch and boots (~6s). By using a longer distance (1.5 meter) proximity sensor, 6 seconds should be able time for the user to approach and position themselves in front of the camera.
 - USB-C connector incorporated for application of power
 - FATP test, debug, and firmware development to be done using USB-C connector
 - Enumerates CLI, tunneling into Raspberry Pi via UART
 - Disabled at end of FATP before packaging
- Vishay VCNL4200 used for Proximity and Ambient Light Sensing
 - Initial research went into the recommended PIR sensors ([slide](#)). Cost and integration when incorporating an ALS grew in complexity. When comparing to a unit such as the VCNL4200 that combines the prox and ALS into a monolithic package ([slide](#)), the VCNL4200 won from a power, cost, and FATP integration point of view.
- A complex power architecture was conceived for powering the IQL-IMX219/FF. Upon further research, the [IQL-IMX219/FF/Eagle](#) was found. It adapts the IQL-IMX219/FF to the Raspberry Pi Zero 2 W's CIS-2 ribbon connection. This alleviated any need for power rail generation, MIPI routing, and added complexity to the design; none of the camera signals touch the camera hat PCB.
 - The [IQL-IMX219/FF/Eagle](#) mounts to the inside of the enclosure, the Camera Hat board goes on top, then the flex connects to the RPi, which gets plugged into the Camera Hat board.
 - This left the necessity to generate a 3V3 rail via a buck converter, due to the case study's requirements.
 - The Raspberry Pi itself generates 3V3 with sufficient current to power the VCNL4200
- Factory testing at a PCBA level is enabled by TPs only on the bottom of the board, which only has the VCNL4200 as the only component, for easy fixturing in a bed of nails

- Everything is shared on GitHub at: https://github.com/auxren/Cam_Hat_Case_Study/
 - [Schematic](#)
 - [KiCad Project](#)
 - [BOM](#)
 - [Power Calculator](#)

Design Considerations

MPN	Manufacturer	Voltage	Current	IO	Range	FOV	Cost	Size	Notes
EKMB1110112	Panasonic	2.3-4V	1uA Sleep 1.9uA Standy 100uA Max Drive	Digital	5m	100° × 90°	\$12.22@500	10.6 x 10.6 x 10.9	Black Rectangle
EKMB110711*	Panasonic	2.3-4V	1uA Sleep 1.9uA Standy 100uA Max Drive	Digita	5m	100° × 90°	\$12.51@500	10.4mm dia 10.9mm tall	White or Black Dome
IRA-S210ST01	Murata	2.7-3.6V	~6uA	Analog	3m	100° × 90°	\$1.3@500	12 x 7 x 5	Cheaper because it doesn't include lease. Analog

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LTR-578ALS-01	Liteon	1.7-3.6V	1uA Standby 95uA Prox 110uA ALS	I2C INT	10cm	-	\$0.42@100	3.9 x 2.36 x 1.35	Combo Prox and ALS Short Distance
VCNL4040M3OE	Vishay	2.5-3.6	300uA + LED Active 460uA Shut Down	I2C	20cm	~40°	\$1.20@500	4 x 2 x 1.1	Combo Prox and ALS Short Distance
VCNL4200	Vishay	2.5-3.6	0.2uA Sleep 213uA Prox 350uA max + LED	I2C	1.5m	~40°	\$1.51@5,000	8 x 3 x 1.8	Combo Prox and ALS